Artificial Intelligence

Lab 2

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Task 1:

Write a function to find mean, median, mode for the given set of numbers in a list.

Task 2: Sets

A **set** is an unordered collection of unique elements in Python. Unlike lists or tuples, sets do not allow duplicate values, which makes them particularly useful when you want to ensure that all elements are distinct. Sets are defined using curly braces `{}` or by using the `set()` function.

Example of a Set:

```
my_set = {1, 2, 3, 4, 5}
```

1. Create Two Sets:
   - Write a Python program that defines two sets of integers.

2. Find Union:
   - Compute the union of the two sets and store the result in another set then print the result.

3. Find Intersection:
   - Compute the intersection of the two sets and store the result in another set then print the result.

Task 3: Word Scramble Game

1. Select a Word:
   - Write a program that selects a random word from a predefined list of words.
   - You can use **random** (`import random`) module for this task.

2. Scramble the Word:
   - Scramble the letters of the selected word randomly. You can use the `random` module for this task.

3. Prompt User to Unscramble:
   - Display the scrambled word to the user and ask them to guess the original word.

4. Provide Feedback:
   - Compare the user's guess with the original word:
     - If the guess is correct, print "Correct! You guessed the word."
     - If the guess is incorrect, print "Incorrect! Try again."

5. Allow Multiple Guesses:
   - Allow the user to keep guessing until they get the correct word.

6. Track Attempts:
   - Keep track of the number of attempts and display it once the user guesses correctly.

Example Output:

```
Scrambled word: tac
Guess the original word: cat
Correct! You guessed the word in 1 attempt.
```

You can use the following list of words:

```
words = [
    "apple", "elephant", "guitar", "banana", "computer",
    "orange", "mountain", "python", "strawberry", "robot",
    "pineapple", "tiger", "airplane", "blueberry", "ocean",
    "book", "raspberry", "chocolate", "watermelon", "galaxy"
]
```

Task 4: Random Username Generator

Create a username generator that produces a random username based on user specifications.

1. Prompt for Username Length:
   - Write a Python program that prompts the user to enter the desired length of the username.

2. Prompt for Username Style:
   - Allow the user to choose the style of the username. The styles can be:
     - Alphabetic: Only letters (both uppercase and lowercase).
     - Alphanumeric: Letters and digits.
     - Mixed: Letters, digits, and special characters (e.g., !, @, #, $).

3. Generate Random Username:
   - Use the random module to generate a username based on the selected style and length.

4. Ensure Style Constraints:
   - If the user chooses "Alphabetic," ensure the username contains only letters.
   - If "Alphanumeric" is selected, include both letters and digits.
   - If "Mixed" is selected, include letters, digits, and special characters.

5. Print Username:
   - Print the generated username to the console.

6. Handle Edge Cases:
   - If the user enters a length that is less than 5 characters, prompt them to enter a length of at least 5 characters.
   - If an invalid style is entered, prompt the user to choose a valid style from the options (this should be case insensitive).

Example Output:

```
Enter desired username length: 10
Enter username style (Alphabetic, Alphanumeric, Mixed): Alphanumeric
Generated Username: f9G7vR2kLQ
```